



# HABIB UNIVERSITY

## Data Structures & Algorithms

CS/CE 102/171 Spring 2025

Instructor: Maria Samad

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Date: April 7<sup>th</sup>, 2025

### Quiz # 3

Time Allowed: 60 minutes

Max Points: 40

Name: \_\_\_\_\_

#### **QUESTION 1: HASHING (15 marks)**

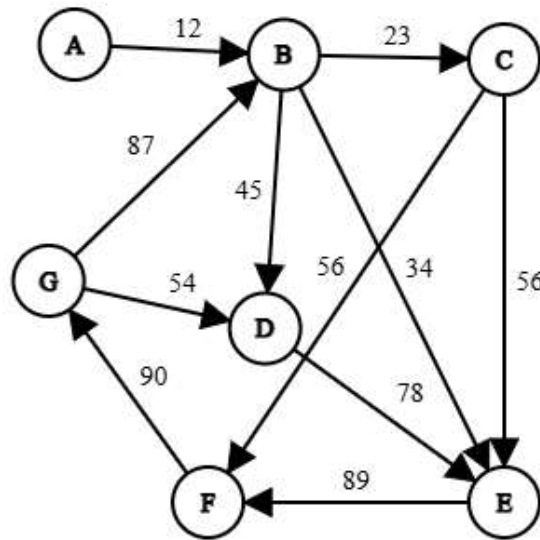
- A list of numbers is given, and you are required to calculate each number's frequency, i.e., how many times it appears in the given list. Write an algorithm using Hashing that populates the hash table in such a way that the number represents the "key" and its frequency represents the "value". The size of hash table is taken as input from the user. Hash function to be used is Divide Modulo, and collisions must be handled using Quadratic Probing. You may use two separate lists to represent keys and values just like you did in the labs. **YOU ARE NOT ALLOWED TO USE ANY ADDITIONAL DATA STRUCTURES, APART FROM THE ONES SPECIFIED IN THE QUESTION.**
- Using the algorithm from part (a), perform hashing on the following list of numbers. Assume size of hash table is 9.
  - [5, 7, 13, 13, 13, 1, 5, 7, 23, 5, 5, 23, 23, 7, 67, 4, 8, 1, 23, 13, 4, 5, 67, 7, 8, 8, 8]



**QUESTION 2: GRAPH TRAVERSAL (10 marks)**

For the given graph, apply the Depth-First Search Traversal Algorithm with the starting vertex = A. Show all the working for each step. Construct the corresponding DFS graph, by showing all the tree edges and non-tree edges. Use different notation to represent each edge, and specify a legend for it. Follow ascending order in choosing vertices. Calculate the total cost of traversal in the end.

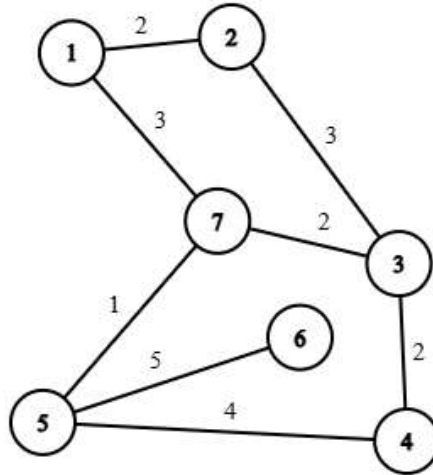
**IF YOU ARE MISSING ANY INTERIM CALCULATIONS, YOU WILL NOT GET ANY MARKS**





**QUESTION 3: SHORTEST PATH (15 marks)**

- a. Run the Dijkstra's Algorithm on the following undirected, weighted graph, to get the shortest path of every vertex from the starting vertex, 1. Specify shortest path of every vertex from the starting vertex, and calculate the cost of this shortest path. Draw the reduced graph after Dijkstra's algorithm has successfully been implemented, showing the shortest paths between all the vertices. Show your complete working. **IF YOU DO NOT SHOW ANY WORKING, YOU WILL NOT GET MARKS FOR IT**



- b. For the same graph, as given in part (a), define the Graph using Adjacency Matrix Representation

